Global Retail Sales

Exploratory Data Analysis

Project Type: End-to-End Portfolio Project
Tools Used: Python (Pandas, NumPy, Matplotlib, Seaborn) · MySQL · Power BI
Dataset: Global Retail Sales Dataset (43 columns)

Problem Statement

Retail businesses operating across multiple regions, brands, and customer segments generate enormous volumes of transactional and operational data — but data alone doesn't create value until it's turned into decisions. Left unexamined, it's easy to assume revenue is concentrated in one market, that price cuts drive sales, or that delivery delays trace back to one obvious cause. This project puts those assumptions to the test using a global retail dataset spanning orders, marketing spend, pricing, logistics, supplier operations, warehouse performance, and customer segments across five regions worldwide.

The analysis is built around four business-critical questions:

1. Sales & Revenue

- Which regions, brands, and product categories actually drive the most revenue — and is the business dangerously concentrated in one market, or is revenue genuinely diversified?
- Does brand leadership hold steady globally, or does it shift from region to region?
- Is growth built on seasonal, festival-driven spikes, or on steady, year-round demand?

2. Marketing & Pricing

- Does higher marketing spend reliably bring in more website traffic, and does more traffic reliably convert into more sales — or does that chain break down somewhere along the way?
- Is the business priced competitively against rivals, and if pricing is already matched, what else is left to compete on?
- Do deeper discounts actually move sales volume, or are they quietly cutting into margin with little to show for it?

3. Logistics & Supply Chain

- Which shipping modes are genuinely the most efficient on both cost and lead time — and does the fastest option also happen to be the cheapest, or is there a trade-off?
- Which warehouses and categories are most exposed to stockouts, and can that be explained by supplier reliability, or does it point to a planning gap?
- Are delivery delays driven by clear factors like congestion, fuel cost, or region — or is this a systemic bottleneck no single variable fully explains?

4. Customer & Discount Behaviour

- Do Retail and Wholesale customers really behave differently in order volume, spend, and discount sensitivity — or do they respond to the business in much the same way?

- Is there a real trade-off between discount depth and effective revenue, or does discounting mostly erode margin without changing buying behaviour?
- Do the two segments convert online at meaningfully different rates, and is that gap large enough to justify separate strategies?

Scope of This Notebook

This notebook covers the Python EDA phase of the project:

- Data loading and initial inspection
- Data quality checks (nulls, duplicates, data types)
- EDA visualisations addressing real business questions
- 5 statistical hypothesis tests (t-test, ANOVA)
- This is one part of a three-tool workflow — see the SQL Analysis and Power BI Dashboard sections below for the rest of the project

Dataset Overview

Property	Detail
Total columns	43
Key domains	Orders, Marketing, Pricing, Logistics, Warehouse, Supplier
Customer segments	Retail, Wholesale
Regions covered	North America, Latin America, Europe, Middle East, Asia Pacific

Notebook Structure

1. Import libraries
2. Load dataset
3. Data inspection (shape, data types, describe, null check)
4. EDA — visualisations
5. Statistical tests — 5 hypothesis tests
6. Summary of key findings

SQL Analysis (MySQL)

Alongside the Python EDA, a set of SQL queries was written and run directly against the cleaned dataset in MySQL Workbench to answer targeted business questions in a query-driven format. This covered:

- Revenue performance by region, brand, and product category, including how brand leadership shifts from one region to another.
- Marketing spend and discount patterns across product categories, brands, and customer segments.

- Order volume trends across months and between Retail and Wholesale segments.
- Pricing analysis, comparing company pricing against competitor pricing and flagging premium-priced orders.
- Warehouse stockout patterns and supplier reliability, broken down by region and product category.
- Shipping mode performance, delivery delays, and logistics cost drivers such as fuel cost, port congestion, and tariffs, viewed by region.

Power BI Dashboard

The findings from the SQL and Python analysis were also brought into a multi-page, interactive Power BI dashboard for ongoing, at-a-glance monitoring. The dashboard is organized into five pages:

- Executive Overview — headline KPIs (total revenue, total orders, average discount rate, stockout rate, supplier reliability, logistics expenditure) alongside global revenue trends.
- Sales & Revenue Analysis — regional, brand, and category-level revenue breakdowns with monthly trend views.
- Marketing & Pricing Analysis — website traffic, conversion rate, marketing spend, and company-vs-competitor price benchmarking.
- Logistics & Supply Chain Analysis — delivery lead time, on-time performance, logistics cost, and carbon emissions by shipping mode.
- Warehouse & Inventory Analysis — stockout incidents, inventory levels, replenishment quantities, and return rates by warehouse, category, and region.